

Trainings ▾

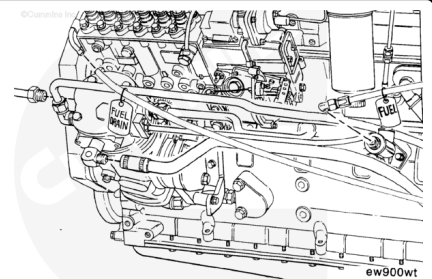
General Information

Installations can vary from OEM to OEM. Use the following steps as a guideline. Reference the OEM manual when necessary.

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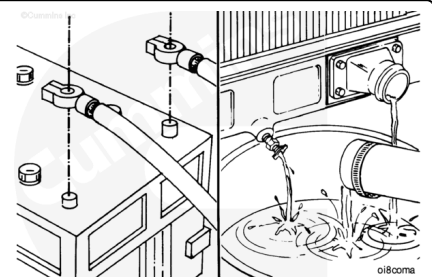
Remove

Place a tag on all hoses, lines, linkage, and electrical connections as they are removed to identify their locations.



⚠ WARNING ⚠

Batteries can emit explosive gases. To avoid personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

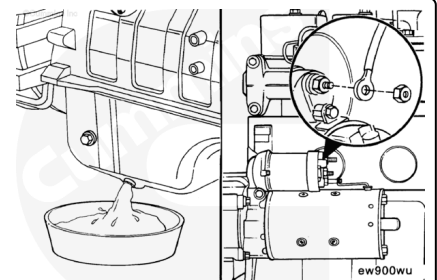
Disconnect the battery cables.

Steam clean the engine. Refer to Procedure 000-009 (</qs3/pubsys2/xml/en/procedures/99/99-000-009.html>).

Drain the engine coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>).

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, or prolonged contact with used engine oil.



⚠ WARNING ⚠

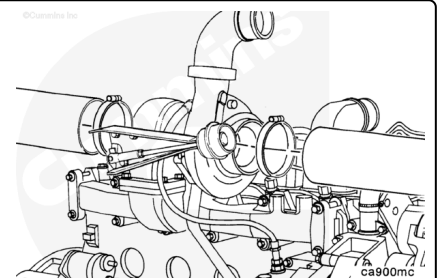
To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/40/40-007-037.html>).

Disconnect the starter cable, engine ground straps, cab- or chassis-to-engine hoses, tubing, electrical wires, and hydraulic lines.

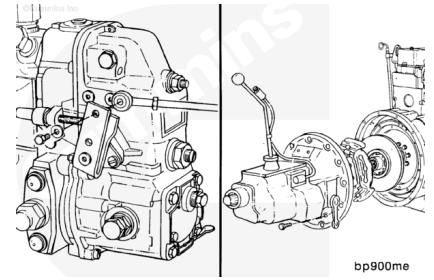
Disconnect the intake and exhaust pipes.

Disconnect all chassis-mounted, engine-driven accessories.



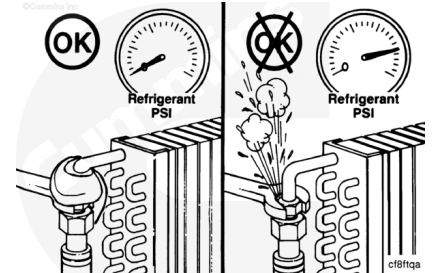
Disconnect the throttle linkage from the fuel injection pump control lever.

Disconnect the drive units from the flywheel.



⚠ WARNING ⚠

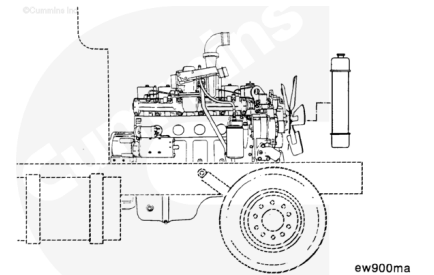
If a liquid refrigerant system (air conditioning) is used, wear eye and face protection, and wrap a cloth around the fittings before removal. Liquid refrigerant can cause serious eye and skin injury.



⚠ CAUTION ⚠

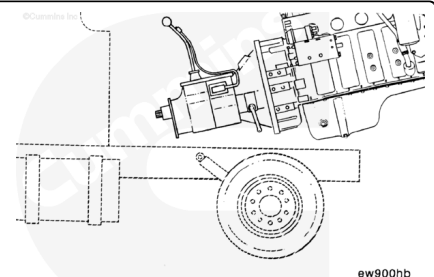
To protect the environment, liquid refrigerant systems must be properly emptied and filled using equipment that prevents the release of refrigerant gas into the atmosphere. Federal law requires capturing and recycling the refrigerant.

Remove all chassis components necessary to remove the engine from the equipment.



⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission safely as an assembly without causing personal injury. For engine weight specifications refer to Procedure 018-015 (</qs3/pubsys2/xml/en/procedures/40/40-018-015-tr.html>), General Engine Specifications in Section V. Refer to the equipment manufacturer's specifications for the transmission weight.



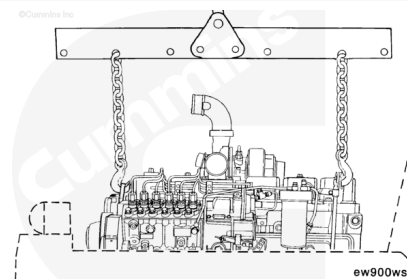
Note : On applications where the rear-engine mounts are attached to the transmission, it is often necessary

to remove the engine and transmission as an assembly. Refer to the OEM service manual.

Use a properly rated hoist and engine-lifting fixture, Part Number 3822512, attached to the engine-mounted lifting brackets to remove the engine.

For engine weight specifications refer to Procedure 018-015 (</qs3/pubsys2/xml/en/procedures/40/40-018-015-tr.html>), General Engine Specifications in Section V.

Note : If the transmission is **not** removed, place a support under the transmission to prevent it from falling.

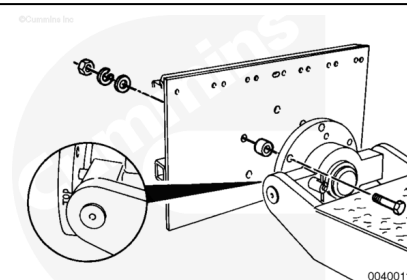


If the engine will be repaired, mount the engine stand adapter plate, part number 3163625, to the exhaust side of the engine.

Note : If equipped, the turbocharger may need to be removed to install the engine stand adapter plate. Refer to Procedure 010-033 (</qs3/pubsys2/xml/en/procedures/40/40-010-033-tr.html>).

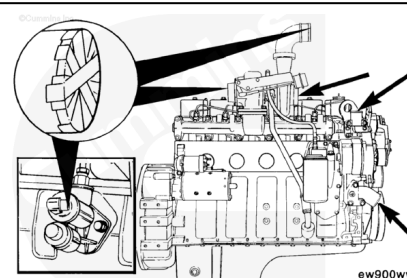
Mount the engine stand adapter plate to the Portable tilt type of engine rebuild stand, Part Number 3375194 or 3375193.

If the engine will be replaced, mount the engine on an appropriate shipping pallet.

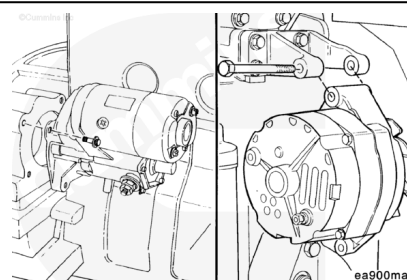


Cover all the engine openings to prevent dirt and debris from entering the engine.

Place the engine on suitable engine-support stands.



If the engine is to be replaced, remove all the remaining accessories, brackets, and drive units to use with the replacement engine.



Last Modified: 10-Apr-2006
